Yingming Zhou

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Education

Stevens Institute of Technology *2025 – Present*
PhD in Computer Science *Provost Doctoral Fellowship*
Capital Normal University *2021 – 2025*
B.S. in Information Management and Information System *2025 Principal's Scholarship*

Research Interests

My research develops reliable LLM systems for software engineering and security. I focus on reinforcement learning with symbolic feedback, access-control policy synthesis, and multi-agent software development.

Selected Publications

(* stands for equal contribution)

[1] RAISE: Reinforcing Access Control Policy Synthesis in LLMs via Symbolic Evaluation
Yingming Zhou, Adarsh Vatsa, William Eiers.
<arXiv preprint, 2026>

[2] PatchHolmes: Agentic Patch Retrieval via Listwise Selection
Guanqun Yang*, Yingming Zhou*, Jiangrui Zheng, Shudong Hao, Xueqing Liu.
<IJCNLP-AACL (Main Conference), 2026>

[3] AutoCedar: An Agentic Framework for Verifier-Guided Access Control Policy Synthesis
Adarsh Vatsa, Sachi Shome, Yingming Zhou, William Eiers.
<arXiv preprint, 2026>

[4] From Reviewers' Lens: Understanding Bug Bounty Report Invalid Reasons with LLMs
Jiangrui Zheng, Yingming Zhou, Ali Abdullah Ahmad, Hanqing Yao, Xueqing Liu.
<Workshop on Secure and Safe AI Agents for Big Data Infrastructure, 2025>

[5] MAO: A Framework for Process Model Generation with Multi-Agent Orchestration
Leilei Lin*, Yumeng Jin*, Yingming Zhou, Wenlong Chen and Chen Qian.
<IEEE Transactions on Services Computing, 2025>

[6] Think-on-Process: Dynamic Process Generation for Collaborative Development of Multi-Agent System
Leilei Lin*, Yingming Zhou* , Wenlong Chen and Chen Qian.
<arXiv, 2024>

Project Experiences

RAISE: Verifier-Guided Access-Control Policy Synthesis [Code]
arXiv preprint, 2026 *May 2026 – Present*

- Built CedarInstruct with 5,800 verified scenarios across 44 domains plus 1,408 organization-specific cases, each paired with executable policy checks.
- Developed RAISE-OC to guide GRPO with failed checks and symbolic counterexamples, improving held-out semantic success by 4.00 points over SFT on Qwen3.5-9B.

ChatDev: Communicative Agents for Software Development
GitHub Open-Source Project — Multi-agent virtual software company framework ***(28K+ stars)*** *Aug 2023 - Nov 2023*

- Designed an adaptive software development process that enables multi-agent workflows to adjust development phases for diverse software requirements.
- Evaluated the framework and iteratively optimized agent prompts based on different development feedback.

Skills

LLM Training & Inference: post-training (SFT, DPO, GRPO with verifier feedback), model distillation; test-time reasoning, tool-integrated reasoning, RAG.
Agent Systems: LLM agents and multi-agent workflows (ChatDev, CAMEL, AutoGen).
Formal Methods: SMT-based symbolic verification (cvc5), Cedar policy synthesis.
ML Tooling: PyTorch, TRL, veRL, vLLM.
Programming & Typesetting: Python, C++, Java, LaTeX.